

Final Project Report

Analyzing and Creating a Soil Database

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1. Introduction

I am a geology major and would like to explore soil factors. Soil is important in geology, agriculture, and land use planning. In geology, it is the product of weathering which can reveal a region's environmental history and determine ground stability for infrastructure. Date Agricultural properties of soil are key to finding good places for crops. Soil data is usually only shown with a few attributes, for many different purposes. For example, a farmer may only care about chemical properties while an engineer cares about drainage and texture of the soil, to ensure stability. This DB aims to standardize these factors in a single model. In addition, finding soil data is difficult and commonly does not always include consistent attributes varying across fields. This can make it harder to compare data and understand how different soil factors affect land use. The USDA Soil Taxonomy allows standardization across global soil samples.

2. Database Functionality and Key Questions

The functionality of this database is relational integrity and temporal tracking. Users can add a new entry with multiple different variables representing different factors of the soil sample taken but can only do so if there is location data and soil_order. The date of samples taken allows comparison of soil over time and looks at age. Chemical and physical properties of each sample are weak entities; these variables could change over time. The same spot can be sampled ten years later and while location will not change, other properties will. Samples can be compared over time, and based on their physical & chemical properties, location, and land

use. Samples in this database can be taken globally, but if a sample is collected from the United States, **state** variable also applies.

- **Do soil orders tend location based globally?**
- **Where is soil for good agriculture?**
- **Do the integers in chemical properties have a relationship with certain variables?**
- **How do soils change temporally?**

Multiple queries will help answer depending on different variables in the soil database. This includes **land_class**, **soil_order**, **drainage_class**, and location. The functionality is relational integrity and temporal tracking. Users can easily add a new soil sample entry with multiple different variables representing different factors, but only if location data and **soil_order** are present.

3. Database Design

The E-R diagram (Fig. 1A) shows the conceptual design of the databases. This database has 5 tables containing various categories of attributes for each soil sample. Every sample entity is a central record of a sampling event. There is a central table named **sample** which is used to connect all entities. Both **chemical_properties** and **physical_properties** are weak entity sets with **sample** as the identifying entity set. Physical and chemical properties cannot exist if the actual sampling event did not occur, each value can be traced back to a specific land use, geological setting, and soil type without redundancy. Composite attributes, **coordinates** and **nutrients** store information that should be viewed together. Derived attribute **sample_age()** will be calculated in the SQL script.

After reducing it to a relational diagram (Fig. 2A) a more clear way to design the database was created. The core table in the database is **sample**; foreign keys used to reference primary keys in other tables. Tables are used to describe various factors of a soil sample chosen. **Sample** includes **date_collected**, **sample_id**, and a foreign keys, which references the primary key from the other tables. Soil property tables are referenced using **sample_id**, with foreign keys of both **sample_id** with **color** is physical and **pH** in chemical.

Table **soil** includes **soil_order** which consists of 12 types of soil, classified by the . U.S. Department of Agriculture (USDA). Each soil type has different properties allowing consistent identification of each soil. Properties vary, including climate, chemical, biological, or physical factors. **Soil_order** acts as a basic overview of the locations' climate and geological history. For example, aridiols are dry, and nutrient poor soils found in desert climates, while mollisols are highly fertile, typically formed in temperate regions (USDA, 1999). **Soil_suborder** is a further refinement of the 12 soil orders. Each has more specific implications for the specific soil e.g. soil moisture and soil development.

Table **location** contains **cname** (country name), **state**, and **coordinates**. This table allows us to see the geospatial spread of the data, or even map it. If users of the database go to find a specific location of a soil that matches specific criteria.

Land_use table is how the land where the soil sample was taken is managed. **land_class** is broad including categories which include agricultural, commercial, open-space, industrial, infrastructure. **Specific_use** is a detailed description of the exact spot the soil was sampled from; e.g. cornfield, or school campus.

Table **physical_properties**. Attributes include color and drain class, describing how the soil looks and behaves physically in terms of water movement and appearance. Soil color helps differentiate soil horizons, which are the result of the dominance of one or more sets of processes over other processes through time (Soil Survey Staff, 1999). Drain class is generally widely accepted as 7 different classes, from excessively - moderately well - very poorly drained. The more excessive the drainage, the faster water will move through the soil (CT ECO, 2010). Very poorly drained soils will have water stuck at the top, not being able to penetrate the surface easily.

Table **chemical_properties** contains chemical measurements including pH, phosphorus, potassium, and cation exchange capacity (CEC). Its purpose is to measure soil fertility and chemical composition for quantitative analysis. Phosphorus and potassium are both major macronutrients for soil, measured in parts per million (ppm) for this database. P, K and pH for a soil depends on what specific crop that is going to be planted. Phosphorus is essential to overall

crop growth, values typically are below 50ppm but ideal for crops is over 50ppm. Potassium is important for water retention and absorption, ideal for most crops with values above 105ppm (Warncke, 2016). A high CEC relates to soils with high organic matter and clay content. A low CEC would be soils made of large grain sizes, like gravel and loamy textures. While a high CEC allows easy exchanging of nutrients, certain soil textures like clay can make it difficult for drainage. CEC is measured in meq/100g. Finally, ideal soil pH for most crops is between 6-7.5, being neutral, slightly basic, or slightly acidic.

4. Data Sources

Fictional data is used for this dataset. For information regarding soil orders and physical properties, USDA documents were referenced. Multiple sources are used for creating fictional chemical properties, to maintain values realistically as possible. For finding specific locations, Isee Network's website *soilexplorer* was used and coordinates where soil orders are globally distributed.

5. Data Analysis

5.1. Database distribution and Standardization (1, 2, 3, 5)

The preliminary analysis (queries 1-3, 5) are set up to see the overall global and soil order distribution in the dataset. All 12 USDA soil orders are represented showing a representation of all soil types. This indicates a diverse representation of this database. Spatial distribution (Fig. 1) shows there are samples across the world in multiple regions. Standardizing latitude and longitude in decimal degrees in table **location**, the database allows simple integration with

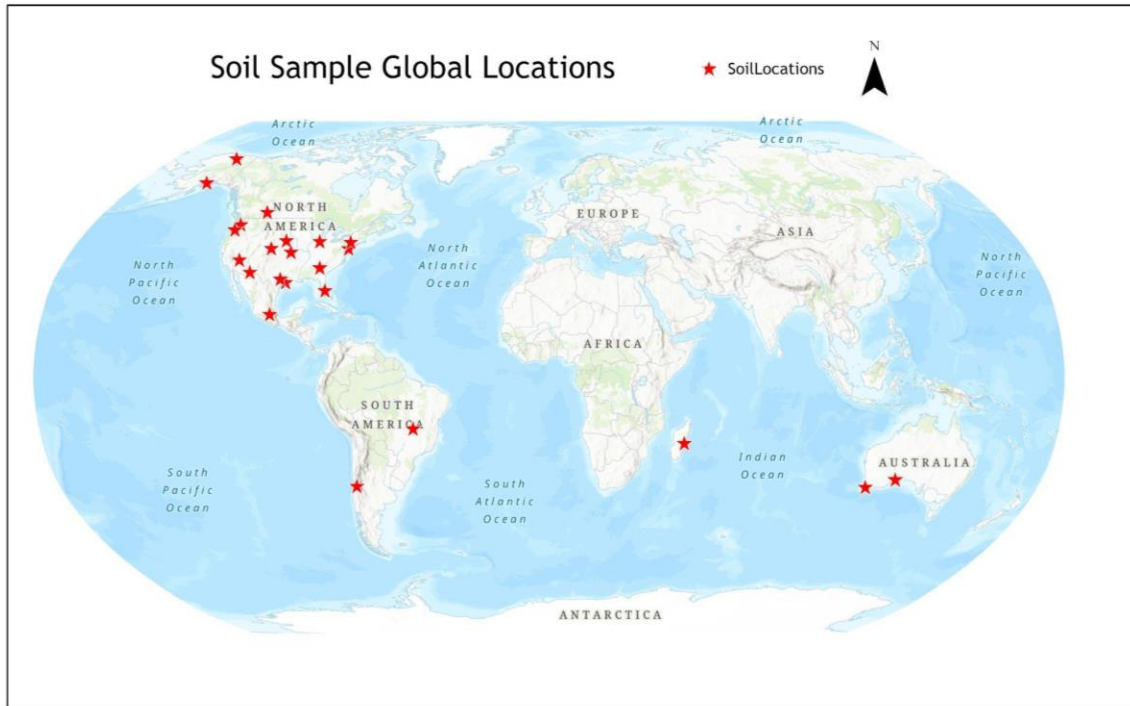


Figure 1. Global Distribution of Soil Samples

mapping to visualize soil distribution.

5.2. Geospatial and Physical Properties

Queries 4, 6, and 7 evaluate whether soil properties are location-based. Figure 2 shows minimal differences in chemical averages between U.S. and non-U.S. samples, indicating soil chemistry is not strongly determined by national geography.

Within the U.S. (Table 1), moderately well-drained and productive soils appear across multiple states and soil orders, suggesting agricultural suitability is not location specific.

Physical properties support this: brown-colored soils are commonly associated with well-drained classes (Table 2), and drainage classes occur across many soil orders (Table 3). Overall, soil behavior is better explained by measurable properties than by location alone.ors driving

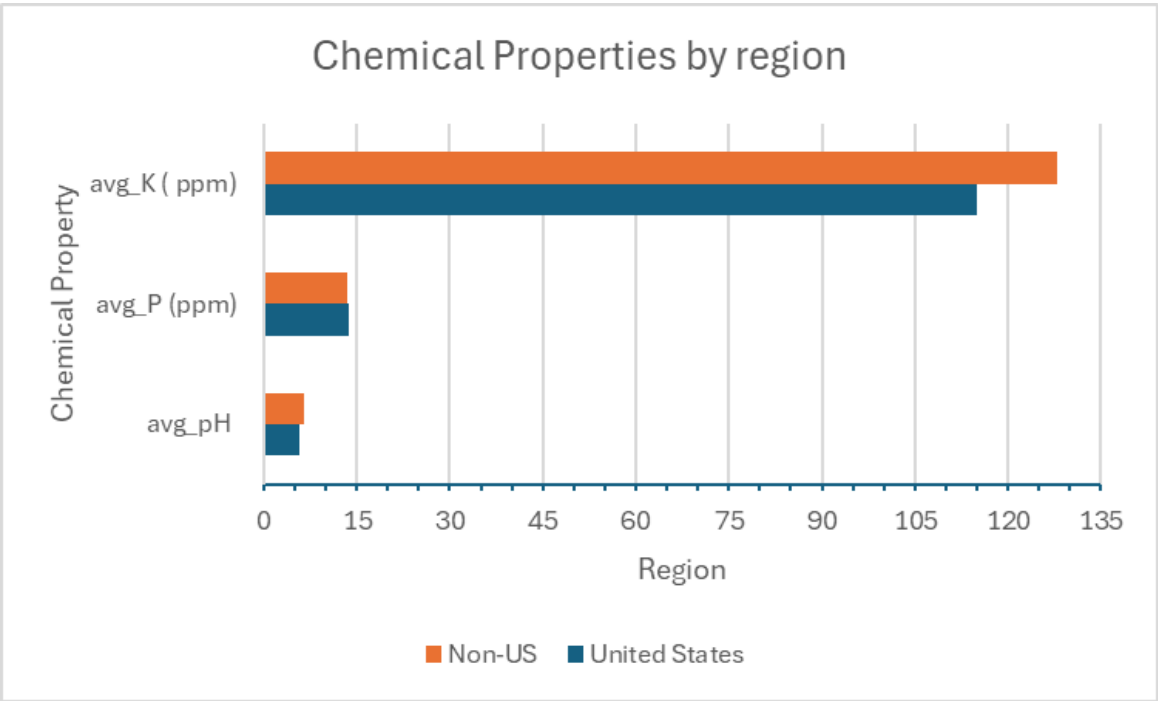


Figure 2. Average Chemical Properties by Region

their values including soil parent material or local climate.

Table 1. Output of Query 8: Which soils are moderately well drained, in the U.S. Only

sample_id	soil_order	state	land_class	specific_use
5	Andisol	Washington	Open Space	Mt. Rainier National Park
6	Mollisol	Kansas	Agriculture	cleared out forest
12	Gelisol	Alaska	Conservation	NULL
23	Vertisol	Illinois	Agriculture	soybean field

Looking at the physical property of color, many samples exhibit 'brown' hues in its name correlates with well-drained classes (Table 2). This indicates more active iron oxidation and high organic matter (USDA 1999). Brown-colored soils are more frequently associated with well-drained conditions (Table 3). Standardizing physical and chemical attributes allows the database to see while soil orders may exist in multiple global locations, its utility for agriculture or engineering is determined by measurable standardized factors rather than location alone. This highlights how observable physical characteristics, like color, can indicate certain underlying soil processes.

Table 2. Output of Query 6: Group color including 'brown' by drain class

color	drain_class	occurrences
reddish brown	excessively drained	2
greyish brown	moderately well drained	1
light brown	moderately well drained	1
dark brown	moderately well drained	1
dark brown	very poorly drained	1
light brown	well drained	1
dark brown	well drained	1
light grayish brown	moderately drained	1
yellowish brown	moderately well drained	1
gray brown	poorly drained	1
orange brown	moderately drained	1
black brown	moderately well drained	1

Table 3. Output of Query 7: Group soil orders by drain class

soil_order	drain_class	sample_count
Alfisol	moderately well drained	1
Alfisol	well drained	1
Andisol	moderately well drained	1
Andisol	poorly drained	1
Aridisol	excessively drained	1
Aridisol	poorly drained	1
Aridisol	well drained	1
Entisol	moderately drained	1
Entisol	very poorly drained	1

Gelisol	moderately well drained	1
Histosol	poorly drained	1
Inceptisol	well drained	2
Mollisol	excessively drained	1
Mollisol	moderately well drained	1
Oxisol	moderately drained	1
Oxisol	poorly drained	1
Spodosol	excessively drained	1
Spodosol	very poorly drained	1
Ultisol	moderately well drained	1
Ultisol	poorly drained	1
Vertisol	moderately well drained	1
Vertisol	very poorly drained	1

5.3. Q3: Chemical Property Analysis (9,10, 12, 15)

To answer the question of chemical integrity and relationships between different variables, multiple queries were run. Queries specifically explore integers in ***chemical_properties*** interactions between each other and other factors.

5.3.1 pH distribution

Query 15 shows soil samples that have above average pH in the dataset specifically. The maximum pH would be 8, so most above average soil sample pH's range from slightly acidic to slightly neutral (Table 4). With soil having these pH values, database design can successfully identify spots in the data where soil's acidity does not hinder agricultural potential.

Table 4. Output of Query 15: List samples that have above average pH (more alkaline)

soil_order	sample_id	pH
Alfisol	3	6.1
Mollisol	6	6.1
Spodosol	21	6.1
Inceptisol	13	6.3
Alfisol	14	6.4
Vertisol	23	6.8
Ultisol	18	6.9
Mollisol	16	7.1
Inceptisol	20	7.4
Vertisol	10	7.6
Aridisol	1	7.9

Aridisol	2	8
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5.3.2 Distribution by soil order

Figure 3 displays a trend connecting soil_class and chemical properties. Phosphorus levels are relatively stable across the dataset, but potassium tends to be higher in orders Entisol and Vertisol orders. High potassium in Vertisols specifically most likely correlates with high clay

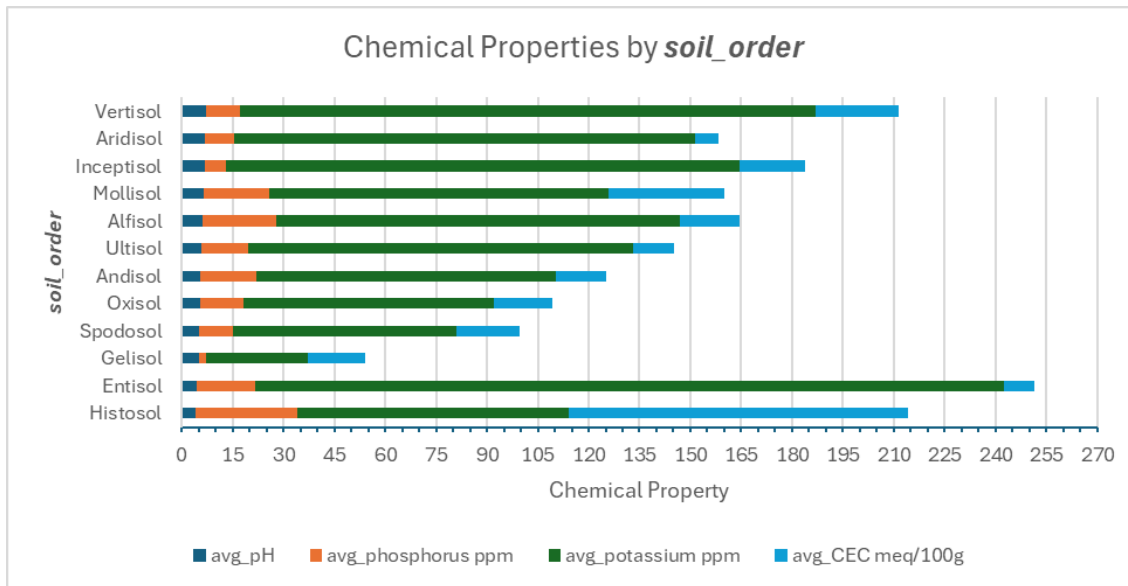


Figure 3. Chemical Properties based on soil_order (Query 10)

content.

5.3.3 Distribution by land_class

Figure 4 groups average chemical properties by land_class. Conservation and Residential have significantly potassium and have much lower CECs. If an area is conservative, it is most likely not to be disturbed by humans while residential areas can be heavily disturbed by building houses. This pattern suggests human activity and land management practices can significantly alter soil chemistry.

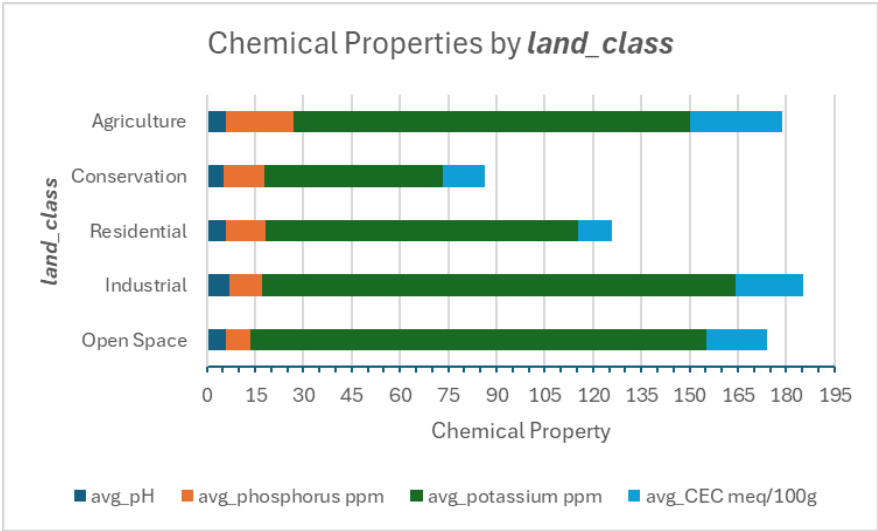


Figure 4. Chemical Properties based on land_class (Query 11)

5.3.3 Distribution by Drain Class

Figure 5 shows chemical averages grouped by drain class. This graph shows there is a larger amount of potassium in the very poorly drained category. This makes sense because the harder it is for soil to drain, the more likely it will stay in the soil and lead to a high CEC. However, increased nutrient retention (CEC) may come at the cost of reduced soil aeration. This can limit plant growth despite having higher nutrient availability.

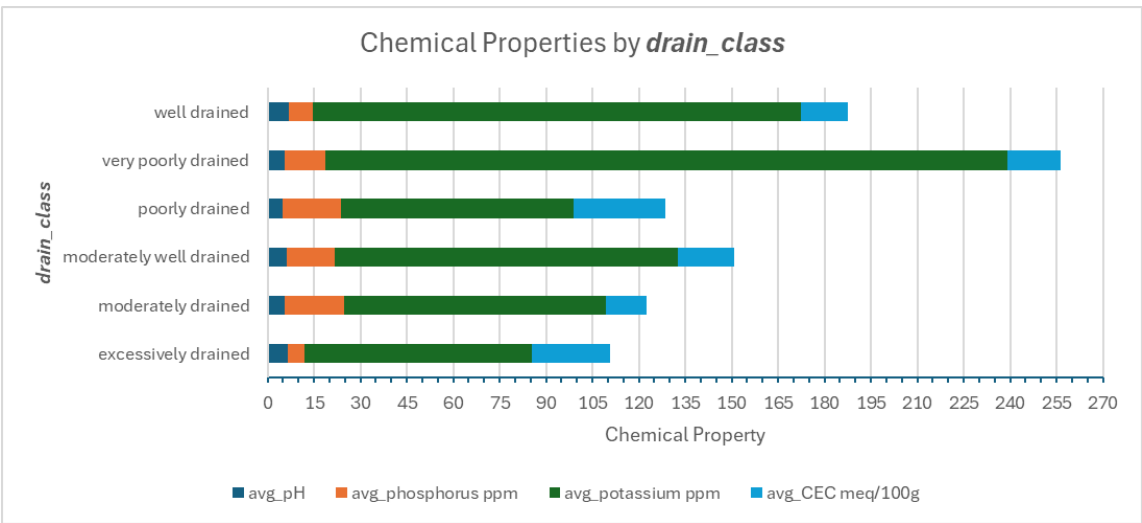


Figure 5. Chemical Properties based on drainage_class (Query 12)

5.4 Temporal Viability and Land-use solutions

Temporal analysis (Figure 6) revealed all soil samples are taken within the past 25 years, generally becoming more frequent with recent years.

This section proves the database's functionality to help find soils ideal for agriculture while also recently collected. First, to find the most productive soils using CEC, the most productive soils specifically in the U.S. were chosen. High CEC does tend to link to high organic matter and clay content. These soils are good for farming; a high amount of clay can lead to challenges for infrastructure. This illustrates the need for the multi-attribute standardization this database provides. These samples were each found in unique states and unique soil_orders, indicating different types of soils and location does not determine if a specific soil_order is good for farming at a specific location. Further, another query refines this further and applies a temporal filter to only select the samples with high nutrients.

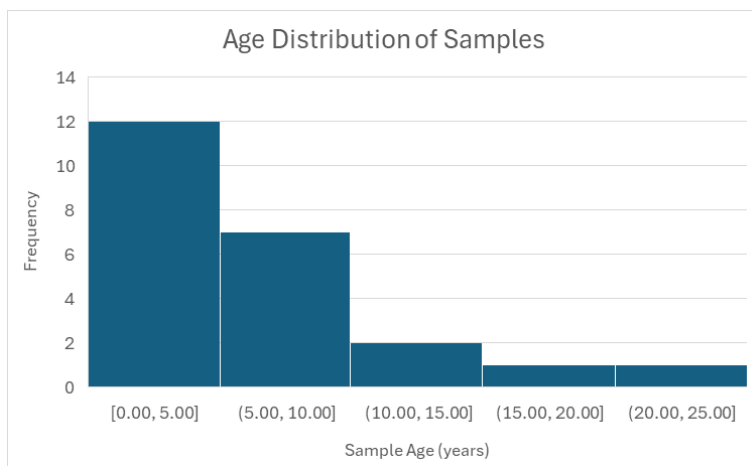


Figure 6. Age of soil sample in years (Query 5)

Conclusion

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Overall, this dataset shows that soil chemical properties vary significantly across soil orders, land use types, and drainage conditions. Agricultural land tends to exhibit higher nutrient concentrations, while soil acidity varies depending on soil classification. These patterns suggest that both natural formation processes and human land use influence soil chemical characteristics. A standardized dataset for soil would be useful for answering these research questions.

References

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7. Appendix

Appendix A

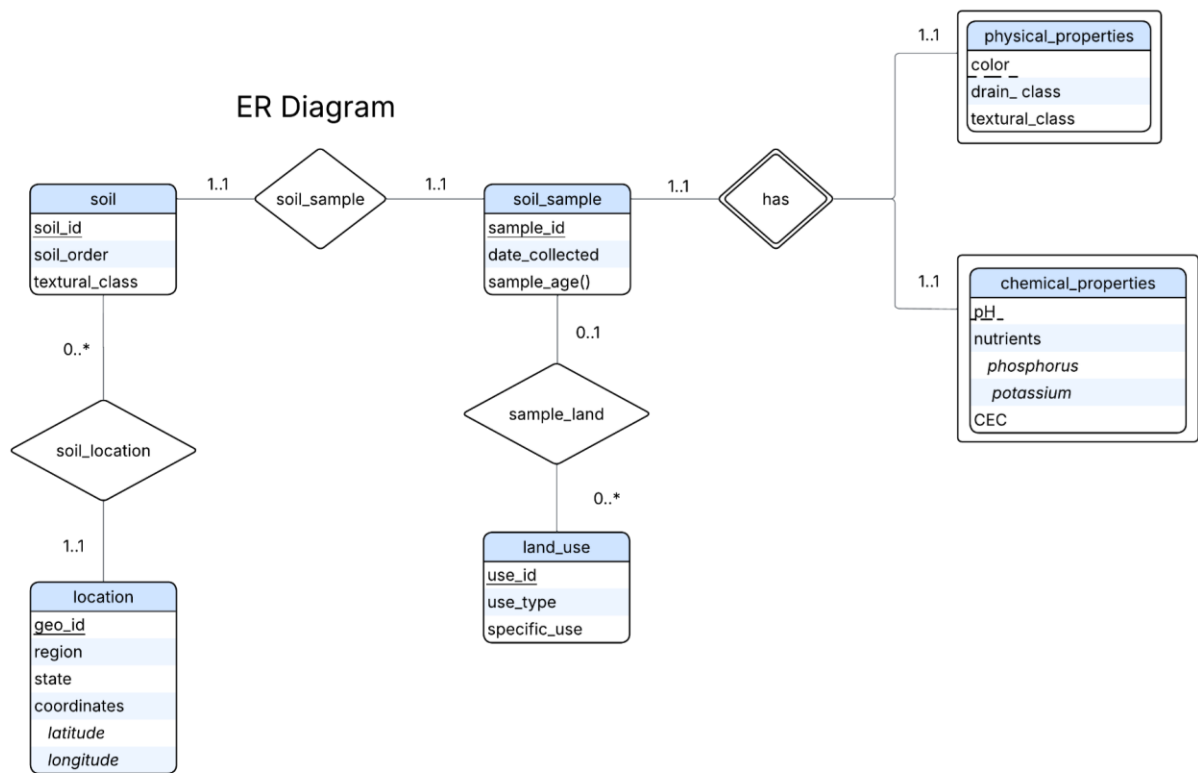


Figure 1A: Entity Relationship Diagram soil samples

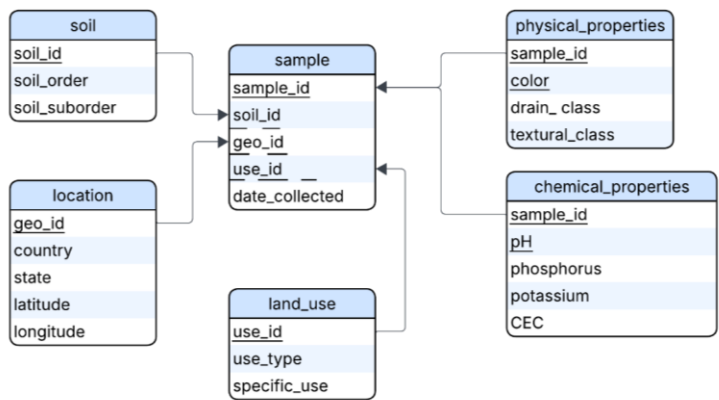


Figure 2A: Relationship Diagram of DB for soil samples